

Sentiment Classification with DistilBERT: Comparing Transfer-Learning Strategies

Problem addressed

Sentiment analysis is an NLP task that automatically determines the sentiment expressed in a text. This project focuses on **binary sentiment classification of movie reviews**, predicting whether a review is positive or negative.

The problem is interesting because movie reviews contain varied vocabulary, sentence structures, opinions, and contextual information. Transformer-based language models can capture these linguistic patterns through pretrained representations, but their performance may depend on how those representations are adapted to a specific task.

Project objective

The objective is to investigate the impact of different transfer-learning strategies when adapting a pretrained DistilBERT model to sentiment classification.

The project compares:

- **Fine-tuning:** the pretrained DistilBERT encoder and classification head are trained together.
- **Frozen encoder:** the pretrained DistilBERT encoder remains fixed and only the classification head is trained.

Research question

How does fine-tuning a pretrained DistilBERT model affect sentiment classification performance compared with freezing the pretrained encoder?

Project workflow

- Select an NLP text-classification task.
- Use the public IMDb dataset from Hugging Face.
- Apply a pretrained `distilbert-base-uncased` model.
- Perform exploratory data analysis and preprocessing.
- Train two transfer-learning configurations.
- Evaluate both approaches using the same test set and metrics.
- Compare and discuss their performance.

Method

Overall strategy

The project uses a pretrained transformer model and compares two transfer-learning strategies under the same experimental conditions.

Common pipeline:

```
IMDb reviews → preprocessing → DistilBERT tokenizer → model training →  
sentiment prediction → evaluation
```

Model : distilbert-base-uncased

- Pretrained transformer language model from Hugging Face.
- Used with a binary sequence-classification head.
- Number of output classes: **2 (Negative or Positive)**
- Framework: **PyTorch + Hugging Face Transformers**
- Hardware: **Google Colab NVIDIA Tesla T4 GPU**

Dataset

The project uses the **IMDb movie review dataset** available through Hugging Face Datasets.

Original dataset:

- 25,000 labeled training reviews
- 25,000 labeled test reviews
- Binary sentiment labels
- Balanced positive/negative classes

The training set was randomly shuffled and split into training and validation subsets using a fixed seed of **42**.

Exploratory Data Analysis

The EDA focused on two aspects:

- **Class distribution:** the dataset contains two balanced sentiment classes.
- **Review length:** review lengths vary substantially, motivating the use of a fixed maximum sequence length during tokenization.

The EDA plots are retained in the accompanying notebook and are used to support the preprocessing decisions.

Data preprocessing

- Load IMDb dataset using Hugging Face Datasets.
- Shuffle the labeled training data using seed 42.
- Split into training and validation sets.
- Load the DistilBERT tokenizer.
- Tokenize review text.
- Truncate sequences to a maximum length of **256 tokens**.
- Apply fixed-length padding.
- Remove the original text column after tokenization.
- Convert the processed datasets to PyTorch-compatible format.

Training setup

Parameter	Value
Model	distilbert-base-uncased
Training samples	4,000
Validation samples	1,000
Test samples	1,000
Maximum sequence length	256
Batch size	16
Learning rate	5×10^{-5}
Epochs	2
Weight decay	0.01
Random seed	42
Precision	FP16
Optimizer	Hugging Face Trainer default
Hardware	NVIDIA Tesla T4

Evaluation metrics

The models are evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Cross-entropy loss

The experiment focuses on a **controlled comparison of transfer-learning strategies** while keeping the main training hyperparameters constant.

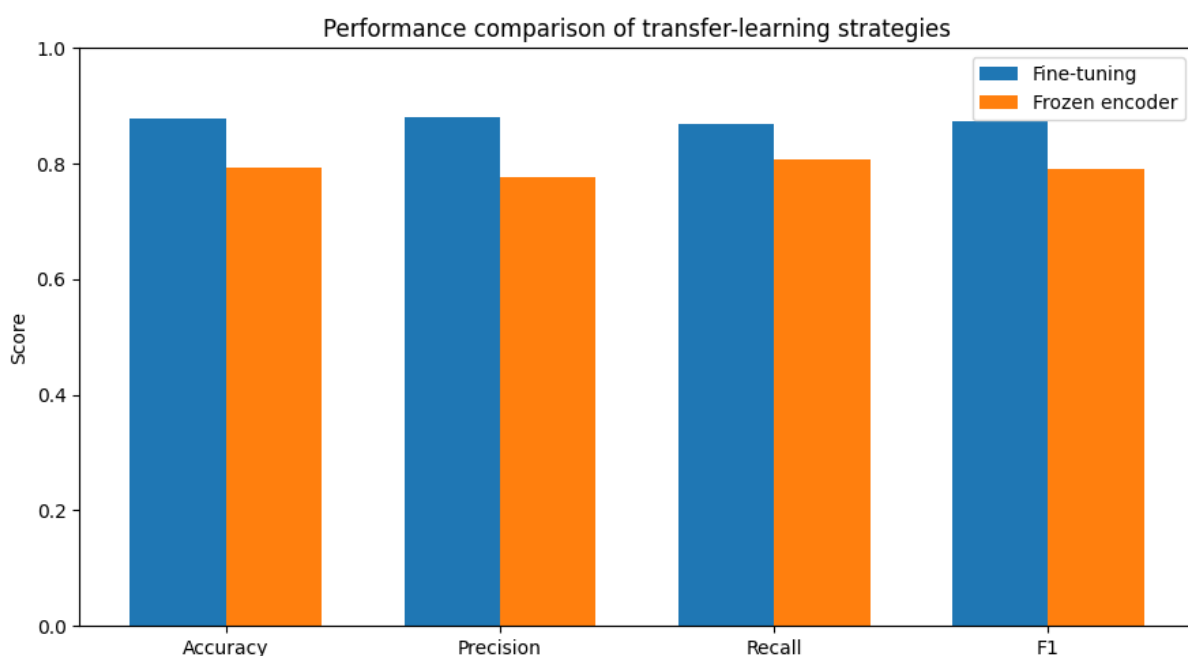
Results

Model performance

The two transfer-learning strategies were evaluated on the same 1,000-example test set. The results are summarized below.

Strategy	Accuracy	Precision	Recall	F1-score
Fine-tuning	87.8%	88.0%	86.9%	87.4%
Frozen encoder	79.3%	77.7%	80.7%	79.2%

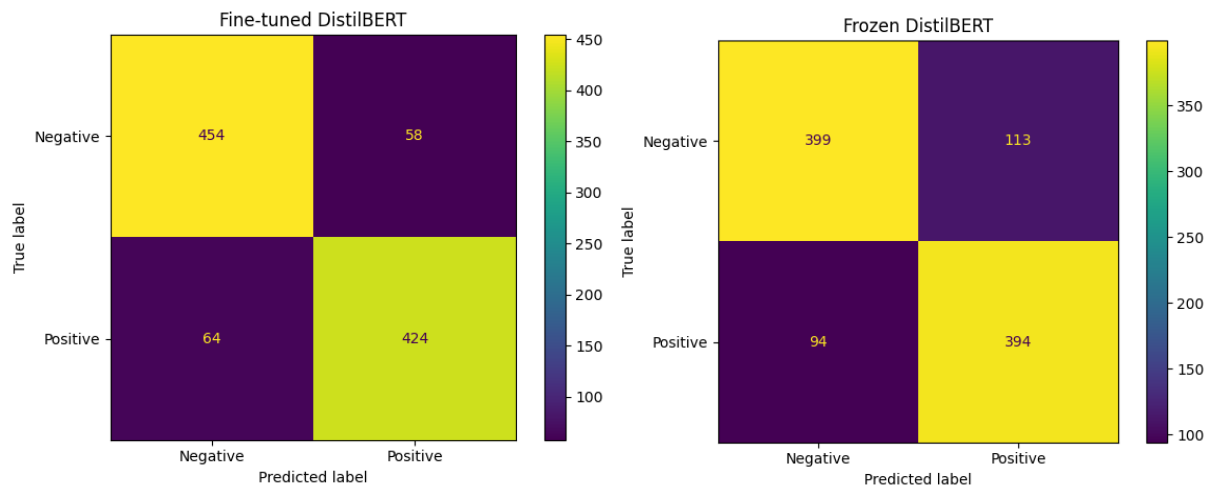
Fine-tuning achieves the highest overall performance, with an accuracy of **87.8%** and an F1-score of **87.4%**. Compared with the frozen encoder, this corresponds to an improvement of **8.5 percentage points in accuracy** and **7.4 percentage points in F1-score**.



The performance difference is also visible in the comparison figure, where fine-tuning consistently achieves higher scores for accuracy, precision, and F1-score. The frozen encoder obtains slightly higher recall, indicating that it identifies a slightly larger proportion of positive examples, but this comes with lower precision.

Confusion matrices

The confusion matrices provide a more detailed view of the classification errors made by each strategy. They show the number of correctly and incorrectly classified positive and negative reviews.



The fine-tuned model produces fewer classification errors overall, which is consistent with its higher accuracy and F1-score. The frozen model remains capable of learning useful sentiment information from the pretrained representations, but its higher number of errors indicates that adapting the encoder to the target task provides additional benefits.

Main observations

- **Fine-tuning achieved 87.8% accuracy**, compared with 79.3% for the frozen encoder.
- Fine-tuning improved **F1-score by 7.4 percentage points**.
- Fine-tuning achieved substantially higher **precision**.
- The frozen encoder achieved slightly higher **recall**.
- Both strategies performed substantially better than random binary classification, showing that the pretrained representations provide useful information for sentiment classification.
- The results support the use of fine-tuning when additional computational resources are available.

Discussion

Interpretation of the results

The results show a clear difference between the two transfer-learning strategies. Fine-tuning achieved **87.8% accuracy and 87.4% F1**, while the frozen encoder achieved **79.3% accuracy and 79.2% F1**. Updating the pretrained DistilBERT weights therefore allowed the model to better adapt its representations to sentiment classification.

The frozen strategy still produced a relatively strong result, with nearly **80% accuracy**. This indicates that the pretrained DistilBERT representations already contain useful linguistic and semantic information for distinguishing positive and negative reviews. However, training only the classification head limits the model's ability to adapt these representations to sentiment-specific patterns.

An interesting difference can be observed in precision and recall. Fine-tuning obtained substantially higher **precision (88.0%)**, while the frozen model achieved slightly higher **recall (80.7%)**. This suggests that the frozen model identified a slightly larger proportion of positive examples, but with more incorrect positive predictions. Fine-tuning provided a better overall balance between precision and recall, resulting in the higher F1 score.

Limitations and improvements

The experiment was conducted on a subset of the IMDB dataset and with a single pretrained architecture, so the results should not be generalized beyond this setting without further evaluation. Additional experiments could use the complete dataset, compare other pretrained models, test partial layer freezing, and evaluate multiple random seeds. A more detailed error analysis could also help identify the types of reviews that remain difficult for both approaches.

Conclusion

Overall, the experiment demonstrates that **fine-tuning a pretrained language model can provide a substantial performance improvement for a specific downstream task**, while freezing the encoder remains a viable approach when preserving the pretrained representations is desirable.